

Purine-rich foods intake and recurrent gout attacks

Objective To examine and quantify the relation between purine intake and the risk of recurrent gout attacks among gout patients. **Methods** The authors conducted a case-crossover study to examine associations of a set of putative risk factors with recurrent gout attacks. Individuals with gout were prospectively recruited and followed online for 1 year. Participants were asked about the following information when experiencing a gout attack: the onset date of the gout attack, clinical symptoms and signs, medications (including antigout medications), and presence of potential risk factors (including daily intake of various purine-containing food items) during the 2-day period prior to the gout attack. The same exposure information was also assessed over 2-day control periods. **Results** This study included 633 participants with gout. Compared with the lowest quintile of total purine intake over a 2-day period, OR of recurrent gout attacks were 1.17, 1.38, 2.21 and 4.76, respectively, with each increasing quintile (p for trend <0.001). The corresponding OR were 1.42, 1.34, 1.77 and 2.41 for increasing quintiles of purine intake from animal sources (p for trend <0.001), and 1.12, 0.99, 1.32 and 1.39 from plant sources ($p=0.04$), respectively. The effect of purine intake persisted across subgroups by sex, use of alcohol, diuretics, allopurinol, NSAIDs and colchicine. **Conclusions** The study findings suggest that acute purine intake increases the risk of recurrent gout attacks by almost fivefold among gout patients. Avoiding or reducing amount of purine-rich foods intake, especially of animal origin, may help reduce the risk of gout attacks.